

LIN7216 – Formal Linguistics

Constructing and interpreting simple sentences

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- Adèle Hénot-Mortier (she/her).
- Thursday 9-11 , Bancroft 3.27.¹
- Office hours Monday 2-3 and Wednesday 10-11 in my office (ArtsOne 1.13) or on Teams.
- QMPlus.
- GitHub (lecture notes only).

¹An additional weekly 1h session will be scheduled starting next week.

Course overview

- **Thematic structure:** each week starting next week we'll focus on a different linguistic object, and analyze it from a formal angle, specifically structure (syntax) and meaning (semantics).
- We will try to start sessions with a short **brainstorming phase**
 during which we will share, list and organize all we already know about the linguistic object(s) at stake.
- Today we'll start with the basics: how sentences are built (**syntax**) and their meaning, computed (**semantics**).
- We'll also briefly cover the construction and meaning of individual words (**morphology**/morphosemantics).

- QMPlus includes **weekly readings**. Core readings (textbook chapters mostly) are **strongly recommended** and directly complement the in-person sessions, while advanced readings (academic papers) are **optional**.
- Having a look at a couple recommended academic papers may help you get a better understanding of **what your assessments should look like** in terms of structure, presentation, language, and argumentation.

Assessments

- Two **identical assessments**: data analysis writeups.
 - Assessment 1 (~ 3000 words) due Wednesday, **November 4th** (middle of reading week).
 - Assessment 2 (~ 3000 words) due Wednesday, **December 11th** (week 12).
- The writeups will be based on the Week 5 and Week 9 sessions.
- During these sessions, we will **analyze unfamiliar linguistic data** together and try to come up with an analysis.
- You will summarize (and potentially improve) the analysis in your assessments, using the **conventions of a linguistics journal paper**: examples, glosses, meaningful sections, scientific



approach, clear argumentation

- Both **form and content** (syntax and semantics!) are relevant to the final mark.

Late penalties and extensions

- It is QMUL policy to penalize late submissions 5 marks per day. In addition, submissions received more than 7 days after the due date will receive a mark of 0.
- Any request for an extension must be made no later than 24h after the due date by submitting an Extenuating Circumstances (EC) claim via MySIS. You can self-certify up to 3 times, or submit a standard claim with appropriate documentation (e.g. a doctor's note).
- If you cannot submit an EC, **it is definitely preferable to submit imperfect work on time than wait and get a 0!** We as instructors cannot make exceptions.

Use of AI

- The use of AI is allowed in this module, **under strict conditions**.
- You **can** use AI as a “personal tutor”, to answer a question about the course, clarify a concept, suggest literature to review.
- You **can** use AI to fix typos and improve style, based on something you wrote yourself.
- You **cannot** use AI to draw syntax trees from scratch (spoiler alert: they may be easy for me to spot) or compute the truth-conditional meaning of an entire sentence.
- You **cannot** use AI to generate entire paragraphs/essays.
- I don't care if your assessments are not conceptually perfect. I will value genuine reasoning, creative ideas, and whether you have integrated the concepts and discussions from class.
- Responsible AI use is not only a matter of intellectual honesty and equity with your peers, it's also a favor you do to your future self, in a world where genuine knowledge is getting scarce!

Brainstorming

The core elements of a sentence

- What's a “good” sentence?
- What typically makes up a sentence?
- How are the elements organized?
- Can you come up with the longest sentence that exists?

Syntax as a data-driven science

- Let's rate the following sentences 🤔

- (1)
 - a. Big black cats purr loudly.
 - b. Colorless green ideas sleep furiously.
 - c. Furiously sleep ideas green colorless.
 - d. Who did you give this book to?
 - e. She (be) working late.
 - f. The car needs washed.
 - g. Who do you think's cat is cute?
 - h. Which problem do you wonder how she can solve?

Grammaticality vs. standard vs. felicity

- As **syntacticians**, we are interested in what is **well-formed**, quite independently of assigned meaning and plausibility.
- Well-formed does not mean **standard**! It means a structure that could be uttered spontaneously in a natural setting, by native speakers of a given language (variety). This corresponds to a **descriptivist** (i.e. non-prescriptivist) approach.
- Well-formed does not mean **probable** either.² Some oddly constructed/convoluted sentences can still be recognized as part of the language, even when nobody has uttered them before.
- Sentences that are well-formed but do not have a “good” meaning in context are **infelicitous**. Felicity is a **pragmatic** concept.

²though sentence probability can influence gradient judgments and reaction times in experimental settings

Grammaticality

- Linguists use **diacritics** to categorize sentences.
 - Ill-formed/ungrammatical sentences are marked *;
 - borderline-grammatical sentences ?, and sentences whose acceptability depends on speakers and/or language variety are %.
 - Infelicitous sentences are #.
- (1)
- | | | |
|----|--|------------------------------|
| a. | Big black cats purr loudly. | Well-formed, makes sense |
| b. | # Colorless green ideas sleep furiously. | Well-formed, makes no sense |
| c. | * Furiously sleep ideas green colorless. | Ill-formed |
| d. | Who did you give this book to? | Well-formed, not “literary” |
| e. | % ³ She (be) working late. | Well-formed in AAE |
| f. | % ⁴ The car needs washed. | Well-formed in e.g. Scotland |
| g. | ?/% Who do you think’s cat is cute | Colloquial and borderline |
| h. | ? Which problem do you wonder how she can solve? | Borderline |

³Given the general English-speaking population.

⁴Given the general English-speaking population.

The job of syntacticians

- Syntacticians work with **judgments**, trying to find the simplest model of sentence-structure that:
 - Predicts the grammatical sentences to be grammatical.
 - Predicts the ungrammatical sentences to be ungrammatical (i.e. does not **overgenerate**).
 - Has the **parametric flexibility** to explain the variable cases, language-internally (dialectal variation, %), and externally (cross-linguistic variation).
- We will work within the **Minimalist** framework, in which syntax builds structure from a small set of simple operations (MERGE), with the resulting structures interpreted at the interfaces with **sound** (Phonological Form, **PF**) and **meaning** (Logical Form, **LF**).
- In the more semantics-y part of the course, we'll see what happens with meanings "at LF", after (some of) the syntactic structure has been built.

The building blocks of sentences

Lexical categories

- The core building blocks of a sentence are words.
- Word can be assigned a **syntactic category**, for instance, noun (**N**). Can you think of others? Adjective (**A**), verb (**V**), determiner (**Det**), preposition (**P**)...
- Adjacent words can be grouped together to form bigger chunks (**constituents**).
- And constituents can also be grouped with other constituents to form even larger constituents!
- For instance, in the sentence *The big cat and the small dog are playing with the ball*, which words seem to form intuitive chunks?

Forming constituents with Merge

- The operation of putting words/constituents together is called **MERGE** in the Minimalist framework.
- Let's try to inductively merge the elements of (2), starting with single words. We'll write two merged (adjacent) elements X and Y [XY].

- (2) The big cat and the small dog are playing with the ball.
- a. The [big cat] and the [small dog] are playing with the ball.
 - b. [The [big cat]] and [the [small dog]] are playing with [the ball].
 - c. [The [big cat]] [and [the [small dog]]] are playing [with [the ball]].
 - d. [[[The [big cat]] [and [the [small dog]]]] are playing [with [the ball]].
 - e. [[[The [big cat]] [and [the [small dog]]]] are [playing [with [the ball]]].
 - f. [[[The [big cat]] [and [the [small dog]]]] [are [playing [with [the ball]]]].
 - g. [[[[The [big cat]] [and [the [small dog]]]] [are [playing [with [the ball]]]]].

Labeling constituents

- In many syntactic frameworks, including vanilla Minimalism, single words or **heads** are labeled according to their syntactic category (N, A, Det, P, V...).
- Constituents are also labeled, according to the word in them that determines their syntactic behavior: the **head** of the constituent.
- A head is said to **project** if it passes its label to a constituent containing it.
- Larger constituents are often **phrases**.⁵, for instance, NPs (e.g. *big cat*), APs (e.g. *black and white*), D(et)P (*the small dog*), PPs (*with the ball*), VPs (*playing with the ball*)...We will sometimes use XP or YP to refer to arbitrary phrases.
- Phrases can contain smaller phrases! Can you identify an example of this configuration in the above?

⁵We'll talk about the difference between phrase and constituent later.

Formally identifying constituents

- So far, we have been forming constituents intuitively, but there are more systematic **tests** to identify them in trickier cases.
 - **Substitution**: constituents can be replaced by single words, e.g. proforms (incl. pronouns), proper names, intransitive verbs...without affecting grammaticality.
 - **Coordination**: constituents can be coordinated with *and*.
 - **Questions and answers**: constituents can be “questioned” and used as short answers.
 - **Emphasis**: constituents can **topicalized** (i.e. fronted, followed by a pause), or **clefted** (*It's ... that* construction in English).
 - **Ellipsis**: constituents can be ellided (i.e. phonologically deleted if a parallel antecedent is available).
- These tests are not fool-proof, but used together offer a fairly robust diagnostic for constituency.

Applying constituency diagnostics

(2') The big cat is playing with the ball.

- *big cat* is a constituent by single-word substitution:
*The **squirrel** is playing with the ball.*
- *The big cat* is a constituent by coordination:
*The big cat **and Fido** are playing with the ball.*
- *The ball* is a constituent by Q&A:
What** is the big cat playing with?–**The ball!
- *With the ball* is a constituent by emphasis:
***It's** with the ball **that** the cat is playing!*
- *Is playing with the ball* is a constituent by substitution:
*The big cat **purrs**.*
- *Playing with the ball* is a constituent by ellipsis:
The big cat is playing with the ball and the small dog ~~is playing with the ball~~ too.

Syntax trees

Drawing trees

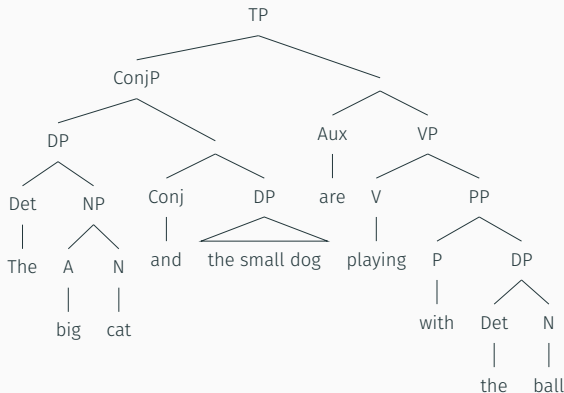
- Below is a parse of the previous sentence, together with partial labeling.⁶
- (2) [[[_{DP}The [_{NP}big cat]] [and [_{DP}the [_{NP}small dog]]]] [are [_{VP}playing [_{PP}with [_{DP}the ball]]]]].
- The [_.] notation used previously to signal merged elements (constituents) is compact but quite heavy 😞
 - Instead, linguists often use trees. If X and Y merge together to form [_{XP} X Y] (where the head X projects), [_{XP} X Y] can be represented as an (unordered) binary branching tree rooted in XP and with daughters X and Y :



⁶Phrase labels usually go on the opening bracket, i.e. to the left of the phrase

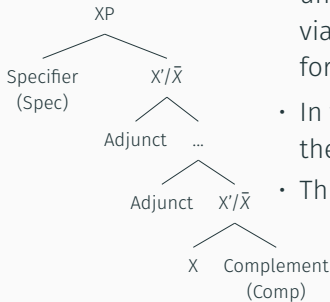
Our first syntax tree

- (2) [[[_{DP}The [_{NP}big cat]] [and [_{DP}the [_{NP}small dog]]]] [are [_{VP}playing [_{PP}with [_{DP}the ball]]]]].



- We use triangles to abstract over obvious/irrelevant structure.
- We consider full standalone sentences (**matrix sentences**) like (2) to be Tense Phrases (TPs).

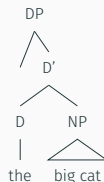
The X-bar schema



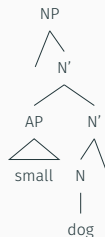
- In the previous tree, some nodes remained unlabeled and the formation of subtrees via MERGE, though intuitive, was not formalized with a single, modular template.
- In the **X-bar schema** any phrase XP follows the same recursive “template” on the left.
- This template distinguishes:
 - optional, stackable **adjuncts** (structurally, sisters of X');
 - from unique **specifier** (daughter of XP and sister of X'), and **complement** (sister of the head X).
- Specifier, adjunct(s) and complement, if non-empty, themselves follow the X-bar schema. These positions will be later shown to fulfil distinct syntactic functions/dependencies/relations.

Applying the X-bar schema

- In *the big cat*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *the* (D), no Spec, Comp is the NP *big cat*.

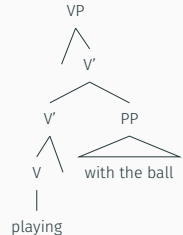


- In *small dog*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *dog* (N), no Spec, one adjunct (the AP *small*), no Comp.

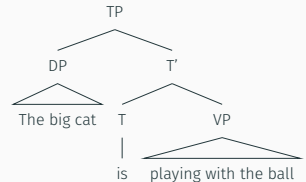


Applying the X-bar schema cont'd

- In *playing with the ball*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *playing* (V), no Spec, one adjunct (arguably, the PP *with the ball*), no Comp.



- In *The big cat is playing with the ball*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *is* (T, present tense), Spec is *the big cat* (the subject DP), no adjunct, Comp is the VP *playing with the ball*.



- **For next week:** draw the complete tree for *The big cat and the small dog are playing with the ball* using X-bar theory.

More X-bar vocabulary

- X' (pronounce X-bar) are called **intermediate projections**, XP **maximal projection**.
- A phrase is a subtree rooted in an XP.
- A constituent is a subtree rooted in an XP, X' , or a head X.
- Is every phrase a constituent? Every constituent a phrase?
- Assuming auxiliaries are in T, *will leave* is a non-phrasal constituent (rooted in T'), which for instance, can undergo ellipsis:

(3) Jo will leave and Al ~~will leave~~ too.

More on projections

- **Lexical** projections like NP or VP have meaning highly dependent on which lexical item (open-class) instantiates the head (compare *jump/give*, or *cat/freedom*).
- **Functional** projections like DP, PP have more deterministic, “functional” meanings, picked from a finite set (closed class, e.g. *a/the*, and *to/from*).
- **Extended** projections are sequences of projections embedding each other. The lowest one is lexical, while the higher ones are functional. PP-DP-NumP⁷-NP is a typical extended projection in

the nominal domain



⁷For Number Phrase

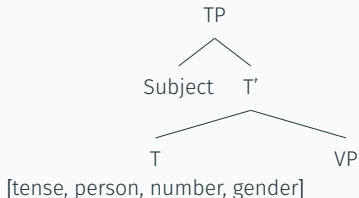
Why is *the cat* a DP and not an NP?

- Whether *the cat* should be treated as a DP (headed by *the*) or an NP (headed by *cat*) is still a matter of heated debate, but **the dominant view is the DP view**. Some evidence for this:
 - Ds **cannot be iterated** (which would be possible assuming they were adjuncts of N): **The this cat*.
 - It makes sense to analyze pronouns like *you* (personal), *this*, *that* (demonstrative) as D-heads with a null complement, because both pronouns and determiners are functional, closed-class elements. And because **pronouns distribute like Det-N sequences**, Det-N is best analyzed as headed by D.
 - It makes sense to analyze **possessors** in Saxon genitives like *Jo's book* as specifiers of null determiners, with a possessee complement: $[_{DP}[_{D'}[_{D}\emptyset][_{NP}\text{book}]]]$.
 - More theory-internally, having functional elements (like Ds) embed lexical elements (like Ns) is **in line with the concept of extended projections**, supported elsewhere in the grammar.⁸

⁸Another example on next slide.

Why are sentences TPs?

- Older frameworks assigned sentences a label S, which was not projected from any of the daughter phrases (typically, an NP subject and a VP). This is **problematic for X-bar theory**.
- The TP solution⁹ does not “choose” between the N and V labels and instead introduces another projection (TP):
 - whose head stores tense/inflectional information (functional information);
 - whose specifier hosts the subject;
 - whose complement is the VP.



⁹You might see “IP” (Inflection Phrase) instead in certain textbooks—roughly the same thing, except IP is more of an umbrella term.

Why are sentences TPs? cont'd

- Treating subjects TP-specifiers is in line with the idea that subjects, *via* **agreement** with the tensed auxiliary or main verb, give the sentence its featural makeup (person, gender, number).
- There's also evidence that sentences "behave" like tensed objects: tense (finite/non-finite) is a feature that clause-embedding verbs are sensitive to. Some select for finite, some for non-finite.

- (4) a. Jo remembers that Al **left**. Finite embedding
 b. Jo arranged for Al **to leave**. Infinitival embedding

- Also notice the parallel with the DP(NP) structure: in a TP(VP) structure, a functional layer (T), embeds a lexical layer (V), to form a (partial¹⁰) extended projection.

¹⁰We'll see that clauses can contain a lot of extra outer functional structure, e.g. CP.

Generative capacity

Infinite outputs from finite means

- We just defined a structure-building system out of a **finite set of labels** (N, D, V, T...), instantiated by a **finite set of lexical items** (*cat, the, purr...*), and following a **fixed tree schema**. The parameters/inputs to this model are finite in number.
- Yet, the system can generate an infinite number of sentences. Can you see why?
 - First, adjuncts can in principle be **iterated *ad infinitum***: *the big furry black cat*.
 - Second (stronger argument), clauses can be **embedded *ad infinitum***: *Jo thinks that Al believes that Sue knows [...] that Ed has left*.
- This comes from the recursivity of the X-bar schema and is important from a cognitive perspective: the finite set of parameters can be stored in the brain as a **Universal Grammar (UG)**, and acquired fairly fast, with the capacity to match the infinity of our outputs 🧠.

Inifinite outputs, really?

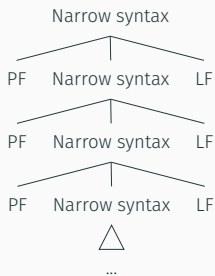
- One could argue the sentences used to motivate the intinity of our outputs get extremely long, and thus unpronouncable/impossible to fully remember!
- It seems to be more about our **general cognitive limitations**, not tied to our internal grammar (UG): working memory, cognitive load, limitations or the articulatory system...
- Assuming these limitations were lifted (“**perfect parser**”), we could imagine ourselves producing the sentences on the previous slide.

Finite means, really?

- What about **open-class categories**? Our lexicon is continually enriched with new nouns, adjectives and verbs, so can we really say our building blocks form a finite set?
- First, there are reasons to think word formation (morphology) follows rules very similar to syntactic rules. In fact, some people argue **morphology is virtually syntax**.
- For instance, one can form a tree for a word like *un-character-istic-al-ly*, which contains a root (*character*), a prefix (*un-*) and multiple suffixes. This **reduces the number of lexical items to posit**, as affixes are a closed-class, and roots often get reused in different words.
- But more fundamentally, the “finite means” of UG do not refer to all the building blocks (roots, words) that *could* be there. Instead, UG ensures that **from any lexicon of fixed, finite size, one can get an infinite number of sentences**.

Interpeting trees

The Minimalist model (aka the Squished Catterpillar¹¹)



- As structure gets built, it gets sent for actual **production** to phonology (PF) and for **interpretation** to semantics (LF) at regular “checkpoints”, called **phases**.
- PF and LF are also called interfaces. Minimalist syntax is often called “narrow” because it delegates a lot of computational work to the interfaces.
- We will now focus on the **LF interface**, which **assigns meaning to trees**.

¹¹©Norvin Richards

Meaning as truth conditions

- The meaning of a sentence are the **conditions under which it is true**. This is the assumption made by **Extensional Semantics**, that we will be using for most of the semester.
- Sentence meaning is evaluated against a situation, materialized as a short **scenario** in linguistic examples. If the scenario satisfies the sentence's truth conditions, the sentence evaluates to the truth-value 1 (and we may flag it as ✓), and if not, the sentence evaluates to 0 (flagged as ✗).
- Some sentence may feel “weird”/uninterpretable in context, but not clearly false. Those will be flagged with #.

(5) Scenario: *Jo is smiling and Al is crying.*

Jo is smiling ✓ Al is smiling ✗

#Jo is smiling too

#Jo is smiling or not smiling

- Just like syntacticians, semanticists use (truth-value) judgments to find the logical interpretation of a sentence that evaluates to 1 exactly in the scenarios where the sentence sounds true. Like syntax, **the ideal semantic system should appeal to a finite set of rules and primitive meanings.**
- We will design a **compositional** system that has these properties: **the meaning of a sentence will be derived from the meaning of its individual constituents and how they (locally) combine together.**
- Semantic composition will be driven by the branchings provided by the syntax tree.

Simplest model: individuals and one-place predicates

- We can build a simple semantic formalism (**metalanguage**) based on **set theory**.
- We posit **individuals** and **sets of individuals** as primitive types of the metalanguage.
- A proper name like *Jo* (a syntactic object from the **object language** that is being interpreted) can be taken to denote the *individual* *Jo*, noted *j* (metalanguage). We write $\llbracket Jo \rrbracket = j$ (“the denotation of the word *Jo* is the individual *j*”).
- Similarly, an intransitive predicate like *smiles* (object language) can be taken to denote the set of individuals who smile, noted $\{x \mid x \text{ smiles} \}$

Deriving the meaning of simple sentences

- A sentence like *Jo smiles* can then be derived by interpreting its two words, and assuming that they semantically combine via set membership relation.

$$\begin{array}{c} \in \\ \swarrow \quad \searrow \\ \llbracket Jo \rrbracket \quad \llbracket smile \rrbracket \end{array} \quad \begin{array}{l} \llbracket Jo \text{ smile} \rrbracket = 1 \text{ iff } \llbracket Jo \rrbracket \in \llbracket smile \rrbracket \\ = 1 \text{ iff } j \in \{x | x \text{ smiles} \} \\ = 1 \text{ iff } j \text{ smiles} \end{array}$$

- We obtained an interpretation of a sentence from the object language (*Jo smiles*) as a truth condition in the metalanguage (true if and only if the individual Jo is a smiling individual).



- This may not seem like much, but shows that from simple lexical entries like $\llbracket Jo \rrbracket = j$ one can *compute* the truth conditions of entire sentences.
- We will see next week sentences with multiple arguments and branching rules and introduce the λ -notation.

Foreshadowing next week: functional meaning

- Suppose negation (expressed by *not* or the prefix *un-*) is a function taking a predicate (set of individuals) and returning its complement set, i.e. all the individuals that are **not** in the original set: $\llbracket \text{not} \rrbracket = \llbracket \text{un-} \rrbracket = S \mapsto \bar{S}$ (“the function mapping any set S to its complement set”).
- One can then compute the denotation of the predicates *not happy/un-happy*, assuming branching semantically amounts to applying a function (negation) to its argument (*happy*).

Function Application


$$\begin{aligned}\llbracket \text{not happy} \rrbracket &= \llbracket \text{un-happy} \rrbracket = \llbracket \text{not} \rrbracket(\llbracket \text{happy} \rrbracket) \\ &= \llbracket \text{not} \rrbracket(\{x \mid x \text{ is happy}\}) \\ &= \{x \mid x \text{ is not happy}\}\end{aligned}$$

$\llbracket \text{not/un-} \rrbracket$ $\llbracket \text{happy} \rrbracket$

- This shows semantics can sometimes apply similarly **above and below the word**.
- Food for thought: reflect on the meanings of *unlockable*, and *unhappier*. Any puzzle/issue?

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